A Project Report on

AUTOMOBILE EMPLOYEE MANAGEMENT SYSTEM

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the project entitled "Automobile Employee Management System" is a bonafide work of **Arnav Sawant(24106015), Rajeev Sangale(24106072), Pushkraj Shirke(24106023), Parth Shinde(24106189)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Abstract

The Automobile Employee Management System is a web-based application developed to efficiently manage employee and operational activities within an automobile organization. The system integrates key modules such as employee management, branch administration, vehicle tracking, sales monitoring, target assignment, and incentive calculation into a centralized and user-friendly platform. The primary objective of this system is to reduce manual workload, minimize errors, and enhance the overall efficiency of organizational processes.

The application enables administrators to store, update, and retrieve employee and sales data seamlessly while providing analytical insights through reports and performance comparisons. This helps management in making informed and strategic decisions. The system is designed using structured methodologies, including Data Flow Diagrams (DFD) and Entity-Relationship (ER) modeling, ensuring a well-organized and scalable architecture.

Furthermore, the use of modern web technologies ensures reliability, security, and ease of use. The system also supports future enhancements such as real-time analytics, mobile application integration, and AI-based predictive analysis. Overall, the Automobile Employee Management System provides a robust and scalable solution for digitizing and optimizing automobile business operations.

TABLE OF CONTENTS

1. Introduction 1
1.1 Motivation 2
1.2 Scope 2
2. Literature Survey 3
3. Limitations of Existing system 4
4. Problem Statement and Objective 5
4.1 Problem Statement 5
4.2 Objective 6
5. Proposed System 7
5.1 Architecture Diagram 7
5.2 System Diagram 9
5.2.1 Data Flow Diagram 9
5.2.2 Use Case Diagram 11
5.3 Methodology 12
6. Experimental Setup 14
6.1 Input Details 14
6.2 Software and Hardware Setup 16
6.2.1 Software Setup 16
6.2.2 Hardware Requirements 16
7. Result and Discussion 17
8. Conclusion and Future Work 28
8.1 Conclusion 28
8.2 Future Scope 29

LIST OF FIGURES

Figure 5.1 3-Tier Architecture Diagram 8
Figure 5.2 DFD Level 0 9
Figure 5.3 DFD Level 1 10
Figure 5.4 Use Case 11
Figure 6.1 ER-Diagram 14
Figure 7.1 Login Page 17
Figure 7.2 Admin Dashboard 18
Figure 7.3 Branch Management and Comparison 19
Figure 7.4 Employee Management and Performance Analysis 21
Figure 7.5 Vehicle Inventory 23
Figure 7.6 All Sales Record 23
Figure 7.7 Managers Dashboard 24
Figure 7.8 Assign Target and Award Incentive 25
Figure 7.9 Employee Dashboard 26
Figure 7.10 Record New Sale 27
Figure 7.11 Sales History of Employee 27

LIST OF TABLES

Table 2.1 Literature SurveyTable 3

ABBREVAION

AI	Artificial Intelligence
ML	Machine Learning
KPI	Key Performance Indicator
DFD	DATA FLOW DIAGRAM
IDE	Integrated Development Environment.
HTML	Hype Text Markup Language
CSS	Cascading Style Sheets.
SQL	Structured Query Language
MDPI	Multidisciplinary Digital Publishing Institute.
AWS	Amazon Web Services
HR	Human Resources
ER	Entity-Relationship

Chapter 1

Introduction

1.Introduction

The **Automobile Employee Management System** is a web-based application designed to manage and analyse employee performance within automobile dealership organizations. In multi-branch environments, tracking employee productivity, sales contributions, and overall branch performance becomes complex when handled manually or through disconnected systems. This application provides a centralized solution to efficiently manage these operations with accuracy and transparency.

Developed using the Django framework, the system integrates employee management, sales tracking, and performance analytics into a single platform. It enables administrators and managers to monitor key metrics such as employee sales, revenue generation, and branch-wise performance through interactive dashboards. Employees can record their sales activities and track their individual progress, while managers can assign targets and evaluate achievements.

By focusing on performance analysis, the system helps organizations identify top performers, improve productivity, and make data-driven decisions. Overall, it enhances operational efficiency and provides valuable insights into both employee and branch-level performance.

1.1 Motivation

The motivation behind developing this system arises from the need to efficiently evaluate employee performance in automobile dealerships. Traditional methods of tracking employee activities and sales performance are often manual, time-consuming, and prone to errors. These limitations make it difficult for management to assess productivity and take timely decisions.

This project aims to provide a structured and automated platform that not only manages employee data but also analyses their performance in real time. The inclusion of branch-level analytics further helps in comparing performance across different locations. Additionally, implementing role-based access ensures secure and organized data handling, motivating the development of this system.

1.2 Scope

The scope of the Automobile Employee Management System includes managing employee records, tracking sales activities, and analysis performance at both employee and branch levels. The system allows administrators to control overall operations, managers to monitor branch performance and assign targets, and employees to record and view their sales performance.

It provides analytical insights such as total sales, revenue contribution, and target achievement, helping in better decision-making. Although the system is currently designed for small to medium-scale organizations, it can be extended in the future with advanced analytics, mobile accessibility, and integration with external systems to support larger enterprise environments.

Chapter 2

Literature Survey/Existing System

2.Literature Survey

Table 2.1 Literature Survey

Sr. No.	Authors & Year	Title of Paper	Methodology Used	Key Findings
1	Anu & Suresh Kumar (2025)	HR Analytics in Automobile Industry	Data-driven HR analytics	Improves employee productivity and decision-making
2	ScienceDirect Study (2024)	Performance in Automotive Dealerships	Regression analysis	Enhances service advisor performance
3	Springer Study (2022)	Predictive Analytics in Sales	Predictive modeling	Improves employee decision-making and sales outcomes
4	Indupriya & Kalaiselvi (2019)	Performance Appraisal in Automobile Companies	Traditional appraisal methods	Moderate effectiveness in evaluation
5	MDPI Study (2019)	Salesperson Performance Evaluation	KPI-based analytics	Identifies key performance indicators
6	Sharma & Sharma (2017)	HR Analytics Framework for Performance	Conceptual HR analytics model	Reduces bias and improves performance evaluation

Chapter 3

Limitations of Existing system

3. Limitations of Existing system

- Most systems focus only on HR analytics or sales performance individually, lacking integration between employee and branch performance.
- Traditional performance appraisal methods are still widely used, which are manual, time-consuming, and prone to human bias.
- Many studies propose conceptual or theoretical models without providing practical implementation or real-world systems.
- Advanced systems using predictive analytics and machine learning require large datasets and high computational resources, making them unsuitable for small and medium organizations.
- Existing solutions often lack real-time performance tracking and dynamic dashboards for better decision-making.
- There is limited support for role-based access control, which is essential for secure and structured management.
- Most systems do not provide user-friendly interfaces or visualization tools, making them difficult to use for non-technical users.
- Some research focuses only on specific roles (e.g., salespersons or service advisors) rather than providing a comprehensive employee performance analysis system.

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

4.1 Problem Statement

To implement a centralized, automated system for automobile dealership organizations that integrates employee data, sales records, and branch performance with real-time tracking, role-based access, and interactive dashboards to enable accurate performance evaluation and efficient decision-making.

4.2 Objective

The main objectives of the Automobile Employee Management System are as follows:

- To develop a centralized web-based system for managing employee, branch, vehicle, and sales data efficiently.
- To implement real-time performance tracking of employees based on sales and revenue metrics.
- To provide branch-wise performance analysis for better comparison and decision-making.
- To enable managers to assign targets and monitor achievements of employees effectively.
- To reduce manual work and improve data accuracy, transparency, and efficiency in operations.
- To build a scalable system that can be extended with advanced analytics and future enhancements.

Chapter 5

Proposed System

5. Proposed System

The proposed **Automobile Employee Management System** is a centralized web-based application designed to efficiently manage employee data and analyze performance at both employee and branch levels. The system integrates sales tracking, employee monitoring, and performance analytics into a single platform.

It provides role-based access for **Admin, Manager, and Employee**, ensuring secure and structured operations. The system focuses on real-time data processing, interactive dashboards, and performance insights to improve decision-making and productivity.

5.1 Architecture Diagram

The **3-Tier Architecture** of the Automobile Employee Management System represents the structured organization of the application into three distinct layers: Presentation Layer, Application Layer, and Data Layer. This architecture ensures better scalability, maintainability, and separation of concerns.

The **Presentation Layer (Frontend)** is the topmost layer that interacts directly with users. It is developed using HTML, Bootstrap, and JavaScript, and is responsible for displaying user interfaces such as dashboards, forms, reports, and charts. This layer allows Admin, Manager, and Employee users to interact with the system easily.

The **Application Layer (Backend)** acts as the core processing unit of the system. It is implemented using the Django framework and handles business logic, request processing, and role-based access control. This layer processes user inputs, performs validations, calculates employee and branch performance, and communicates with the database.

The **Data Layer (Database)** is the bottom layer responsible for storing and managing all system data. It uses an SQLite database to maintain records such as employees, vehicles, sales transactions, targets, and incentives. This layer ensures data integrity and supports efficient data retrieval.

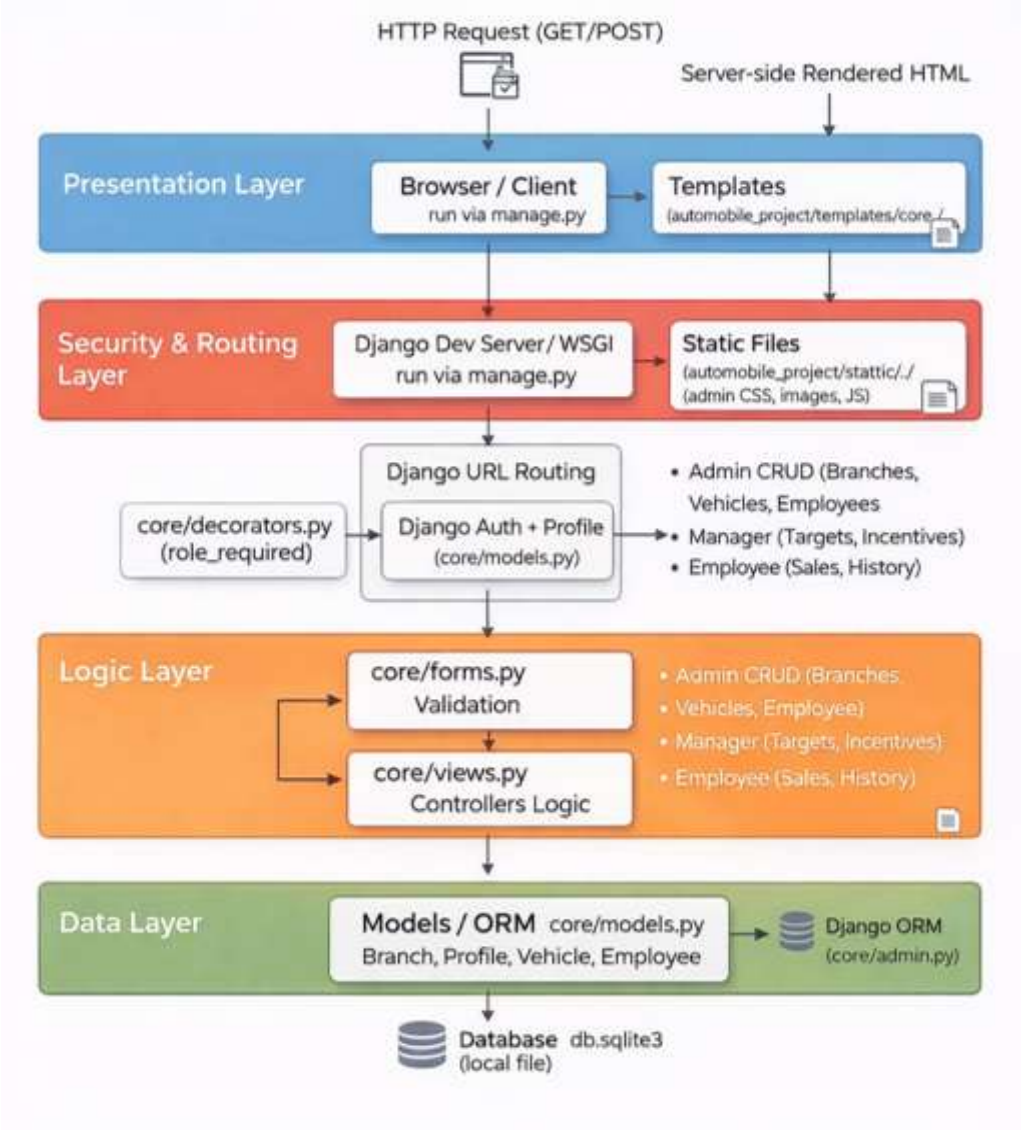

Figure 5.1 3-Tier Architecture Diagram

Overall, the 3-tier architecture enhances system performance, security, and flexibility by separating user interface, business logic, and data management into independent layers.

5.2 System Diagram

5.2.1 Data Flow Diagram

The **DFD Level 0**, or Context Diagram, provides a high-level view of the Automobile Employee Management System by showing its interaction with external entities. The entire system is represented as a single process that receives inputs from users (Admin, Manager, and Employee) such as employee data, sales details, and targets.

The system processes this data, stores it in the database, and generates outputs in the form of dashboards, reports, and performance analytics. This diagram focuses only on overall data flow and does not include internal processes, making it a simplified representation of the system.

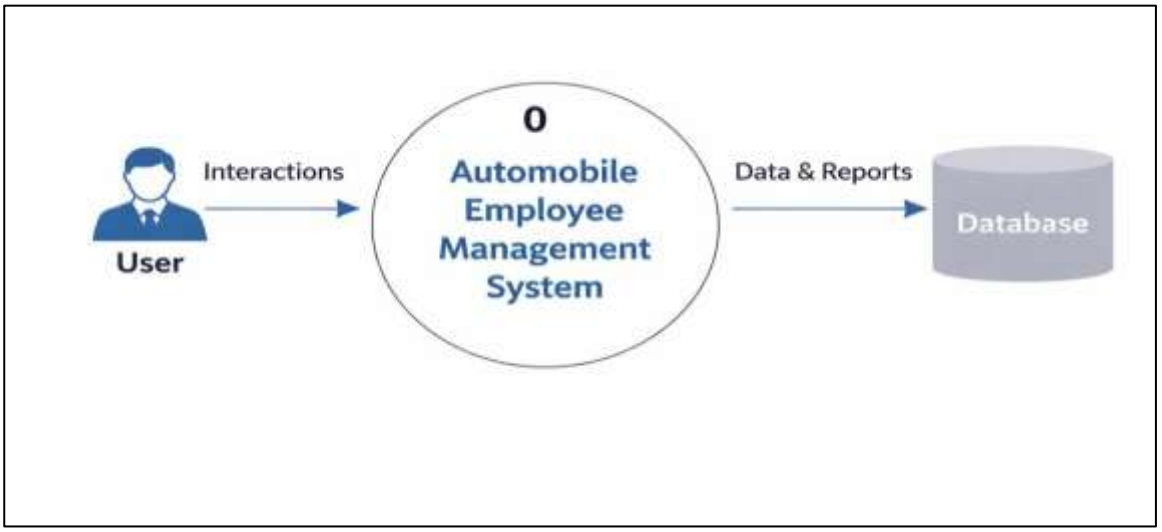

Figure 5.2 DFD Level 0

The **DFD Level 1** provides a detailed representation of the Automobile Employee Management System by breaking down the main system into multiple sub-processes. It shows how data flows between different processes, users, and the database.

In this diagram, the system is divided into key processes such as **Manage Employees**, **Record Sales**, **Assign Targets**, and **Performance Analysis**. The **Admin** interacts with the system to manage employee and branch data, while the **Manager** assigns targets and monitors performance. The **Employee** records sales and views personal performance.

All these processes interact with a central **database**, where data such as employee details, sales records, and targets are stored and retrieved. The **Performance Analysis** process uses this data to generate outputs like reports, dashboards, and analytics.

Unlike DFD Level 0, this diagram provides a deeper understanding of the internal working of the system by showing how different components are interconnected. It helps in visualizing the flow of data and the functioning of each module within the system.

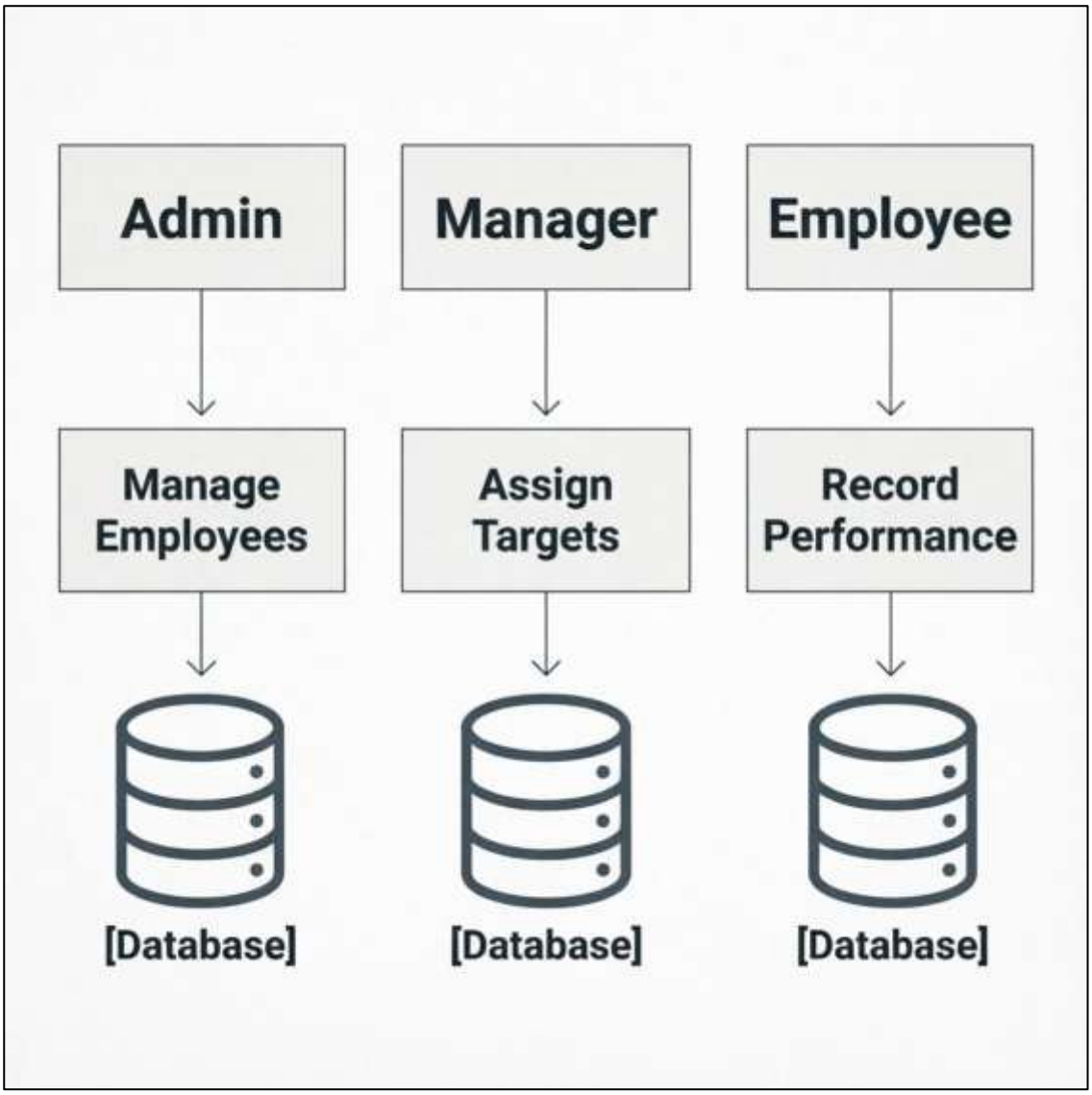

Figure 5.3 DFD Level 1

5.2.2 Use Case Diagram

The **Use Case Diagram** of the Automobile Employee Management System represents the interaction between different users (actors) and the system functionalities. It provides a high-level overview of how various roles utilize the system to perform specific tasks.

The system consists of three primary actors: **Admin, Manager, and Employee**, each having distinct responsibilities and access levels. The **Admin** has complete control over the system and can manage branches, employees, and vehicles, as well as view overall reports and analytics. The **Manager** is responsible for monitoring branch performance, assigning sales targets, and evaluating employee progress. The **Employee** interacts with the system by recording sales transactions and viewing their individual performance dashboards.

Each actor is connected to specific use cases that define the operations they can perform. These include activities such as managing data, assigning targets, recording sales, and viewing performance reports. The diagram helps in understanding system functionality, user roles, and their interactions in a clear and structured manner.

Overall, the Use Case Diagram simplifies the visualization of system behavior and ensures that all user requirements are properly captured during system design.

Figure 5.4 Use Case

5.3 Methodology

The development of the Automobile Employee Management System follows the **Agile Development Methodology**, which emphasizes iterative development, flexibility, and continuous improvement. This approach ensures that the system is developed in phases with regular feedback and enhancements.

1. Requirement Analysis

In this phase, the system requirements are identified and analyzed. The key functionalities such as role-based access (Admin, Manager, Employee), employee management, sales tracking, target assignment, and performance analytics are defined. User needs are carefully studied to ensure the system meets real-world dealership requirements.

2. System Design

This phase involves designing the overall architecture and structure of the system. The database schema is created, defining entities such as Branch, Employee, Vehicle, Sale, and Target. Additionally, system architecture (3-tier architecture), user interface layouts, and data flow diagrams are designed to provide a clear blueprint for development.

3. Development

The actual implementation of the system is carried out in this phase. The backend is developed using the Django framework to handle business logic, while the frontend is built using HTML, Bootstrap, and JavaScript for an interactive user interface. Features such as dashboards, forms, and role-based navigation are implemented.

4. Testing

Testing ensures that the system functions correctly and meets the specified requirements. Unit testing is performed to verify individual components, while integration testing ensures smooth interaction between modules. Errors and bugs are identified and resolved to improve system reliability.

5. Deployment

In this phase, the system is deployed on a local server for execution and demonstration. The application is configured, and the database is set up to ensure proper functioning. Users can access the system through a web browser.

6. Maintenance

After deployment, the system enters the maintenance phase, where continuous monitoring and improvements are carried out. Bug fixes, performance enhancements, and feature updates are implemented to ensure the system remains efficient and up-to-date.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Input Details

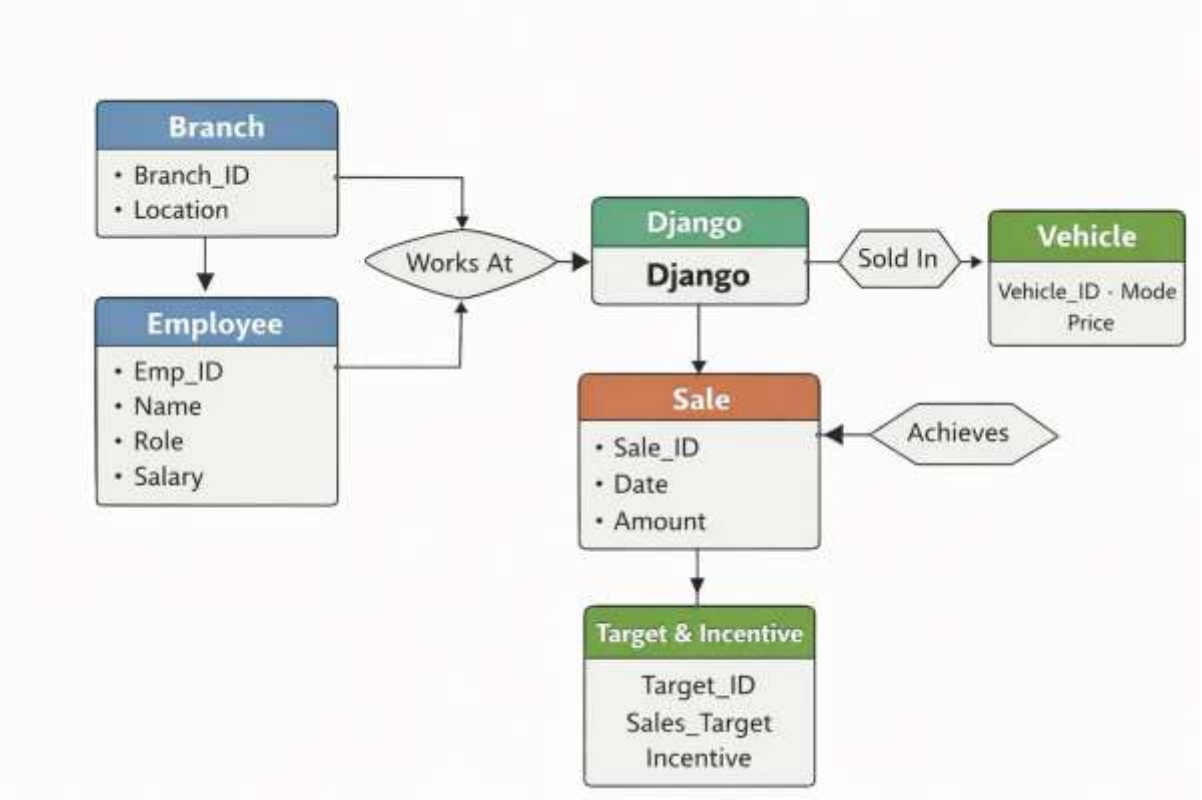

Figure 6.1 ER-Diagram

The **Entity-Relationship (ER) Diagram** represents the structure of the database used in the Automobile Employee Management System. It illustrates the main entities, their attributes, and the relationships between them, which define how data is stored and managed within the system.

The primary entities identified in the system are:

- **Branch:**
  Contains details of different branches such as *Branch_ID* and *Location*. Each branch acts as a central unit where employees are assigned and operations are performed.

- **Employee:**
  Stores information about employees including *Emp_ID, Name, Role,* and *Salary*. Each employee is associated with a specific branch.

- **Vehicle:**
  Includes vehicle-related data such as *Vehicle_ID, Model,* and *Price*. These vehicles are available for sale within the system.

- **Sale:**
  Represents transaction records with attributes like *Sale_ID, Date,* and *Amount*. This entity plays a key role in tracking business performance.

- **Target & Incentive:**
  Stores performance-related data including *Target_ID, Sales_Target,* and *Incentive*, which are used to evaluate employee performance.

Relationships:

- **WorksAt:**
  Defines the relationship between *Employee* and *Branch*, indicating that employees work at a specific branch.

- **SoldIn:**
  Represents the relationship between *Vehicle* and *Sale*, showing that vehicles are sold through sales transactions.

- **Achieves:**
  Connects *Sale* with *Target & Incentive*, indicating that sales contribute toward achieving targets and incentives.

6.2 Software and Hardware Setup

6.2.1 Software Setup

The development and implementation of the Automobile Employee Management System require the following software components:

- **Operating System:** Windows 10/11 (or Linux/MacOS)
- **Programming Language:** Python 3.x
- **Framework:** Django 4.2
- **Database:** SQLite3
- **Frontend Technologies:** HTML5, CSS3, Bootstrap 5, JavaScript
- **Libraries/Tools:**
  - Django Humanize (for formatting data)
  - Chart.js (for data visualization and graphs)
- **IDE/Code Editor:** Visual Studio Code
- **Web Browser:** Google Chrome / Microsoft Edge
- **Version Control (Optional):** Git

6.2.2 Hardware Requirements

The system can run efficiently on standard hardware with the following specifications:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 500 MB free disk space
- **System Type:** 64-bit system
- **Network:** Internet connection (for updates and dependencies)

Chapter 7

Result and Discussion

7. Result and Discussion

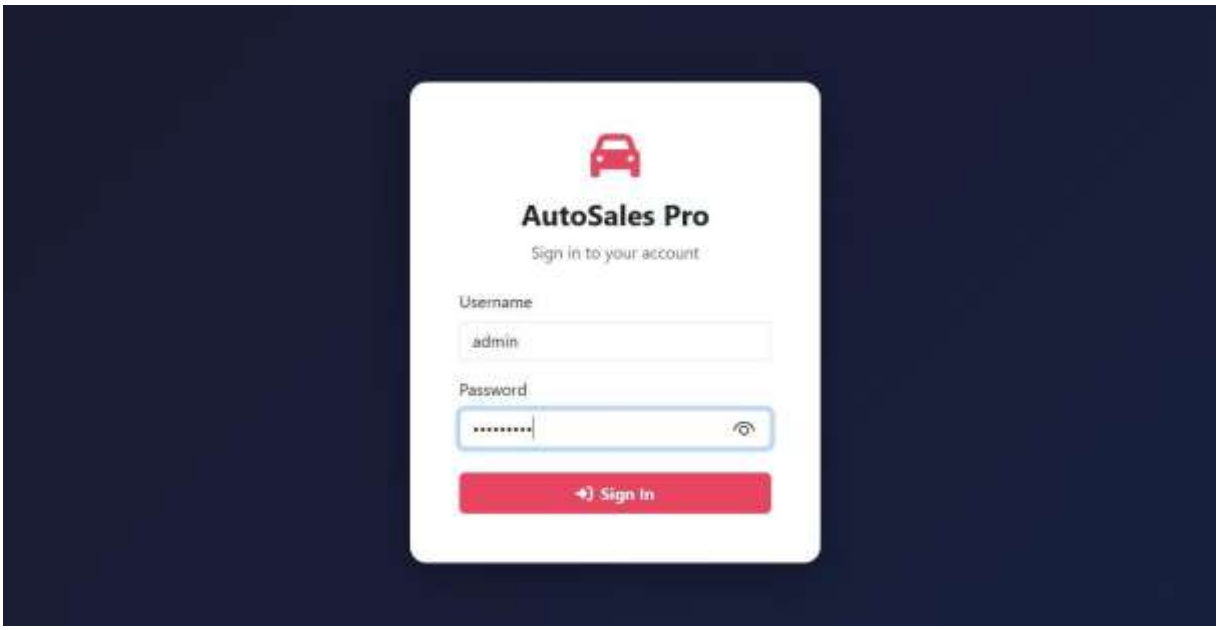

Figure 7.1 Login Page

The **Login Page** of the Automobile Sales Management System (Auto-Sales Pro). It serves as the entry point for all users accessing the system. The page implements secure authentication to ensure only authorized personnel can access the application.

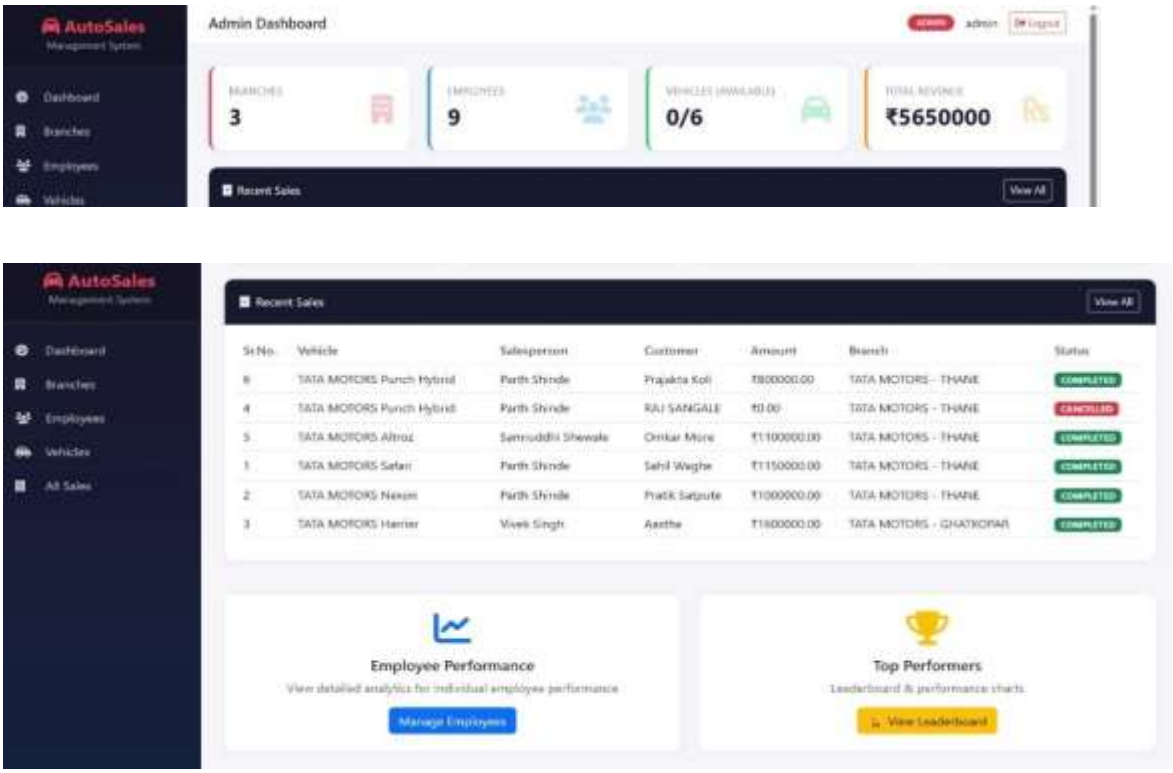

Figure 7.2 Admin Dashboard

The Admin Dashboard is the central control panel providing administrators with a real-time, comprehensive overview of the entire dealership network. It delivers immediate insights into core business metrics, recent transactions, and workforce analytics.

- **Sidebar Navigation:** Quick access to manage core system modules: Branches, Employees, Vehicles, and All Sales.
- **KPI Summary Cards:** Real-time metrics displaying total active branches, registered employees, current vehicle inventory (available vs. total), and total generated revenue.
- **Recent Sales Table:** A system-wide log of the latest transactions showing vehicle, salesperson, customer, amount, branch, and color-coded status (e.g., Completed, Cancelled).
- **Quick Action Modules:** Direct shortcuts to detailed "Employee Performance" analytics and the "Top Performers" leaderboard.
- **Header Controls:** Top navigation displaying the active user role (ADMIN), username, and a secure logout button.

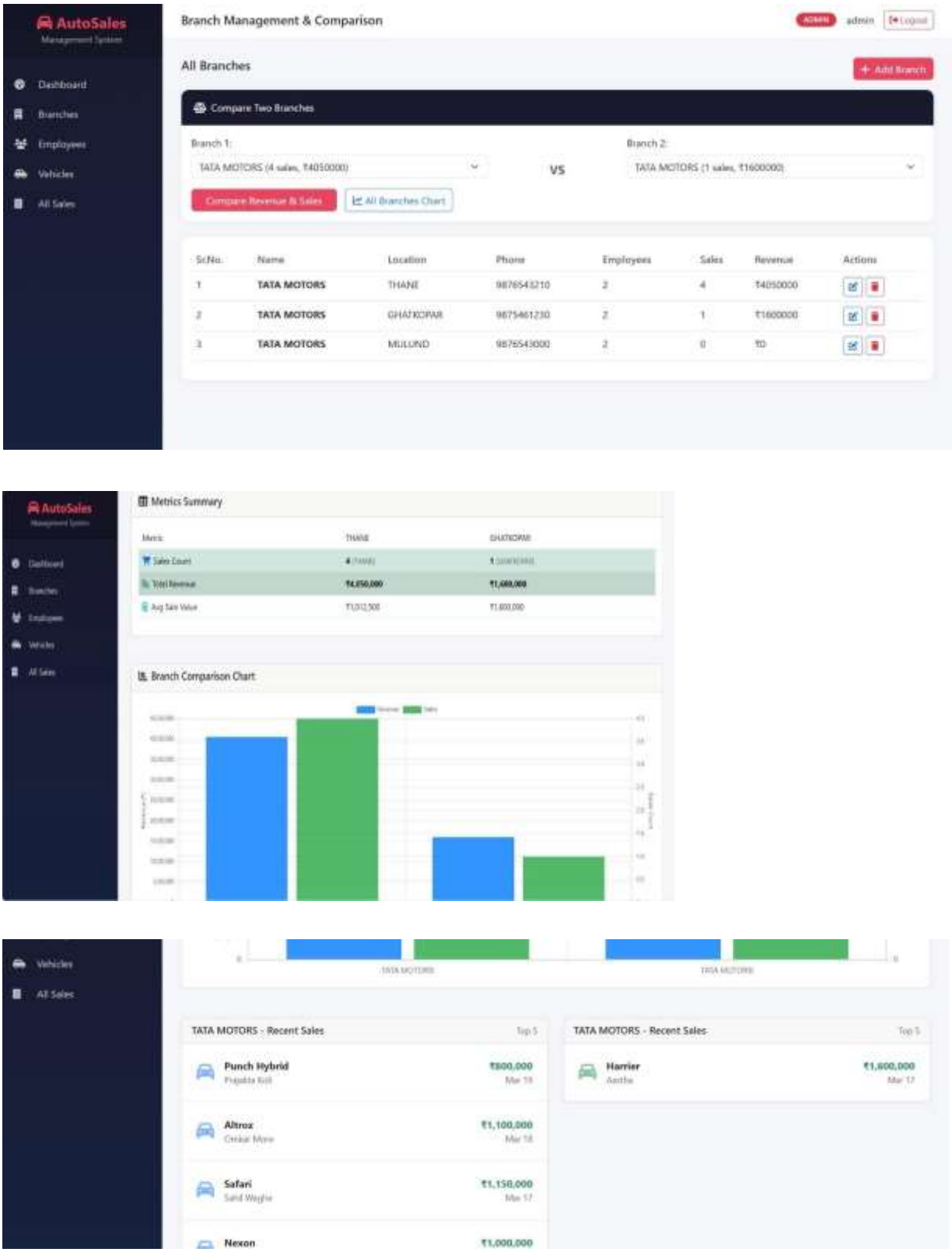

Figure 7.3 Branch Management and Comparison

The Branch Management and Comparison page enables the administrator to manage branch details and compare the performance of different branches in one place. It combines branch records with analytical tools to help evaluate sales, revenue, and overall branch productivity.

Key Features:

- **Branch Listing:** Displays all branches with details such as branch name, location, phone number, number of employees, sales count, and total revenue.
- **Branch Comparison:** Allows the admin to select two branches and compare their sales and revenue performance.
- **Metrics Summary:** Provides a clear summary of branch-wise metrics such as sales count, total revenue, and average sale value.
- **Comparison Chart:** Presents branch performance visually through charts for easy understanding and analysis.
- **Recent Sales Overview:** Shows the latest sales made by selected branches for quick review.
- **Top Vehicle Sales:** Highlights the best-selling vehicles in each branch.
- **Branch Actions:** Includes options to add, edit, and delete branch records.
- **Decision Support:** Helps management identify stronger and weaker branches for better planning and performance improvement.

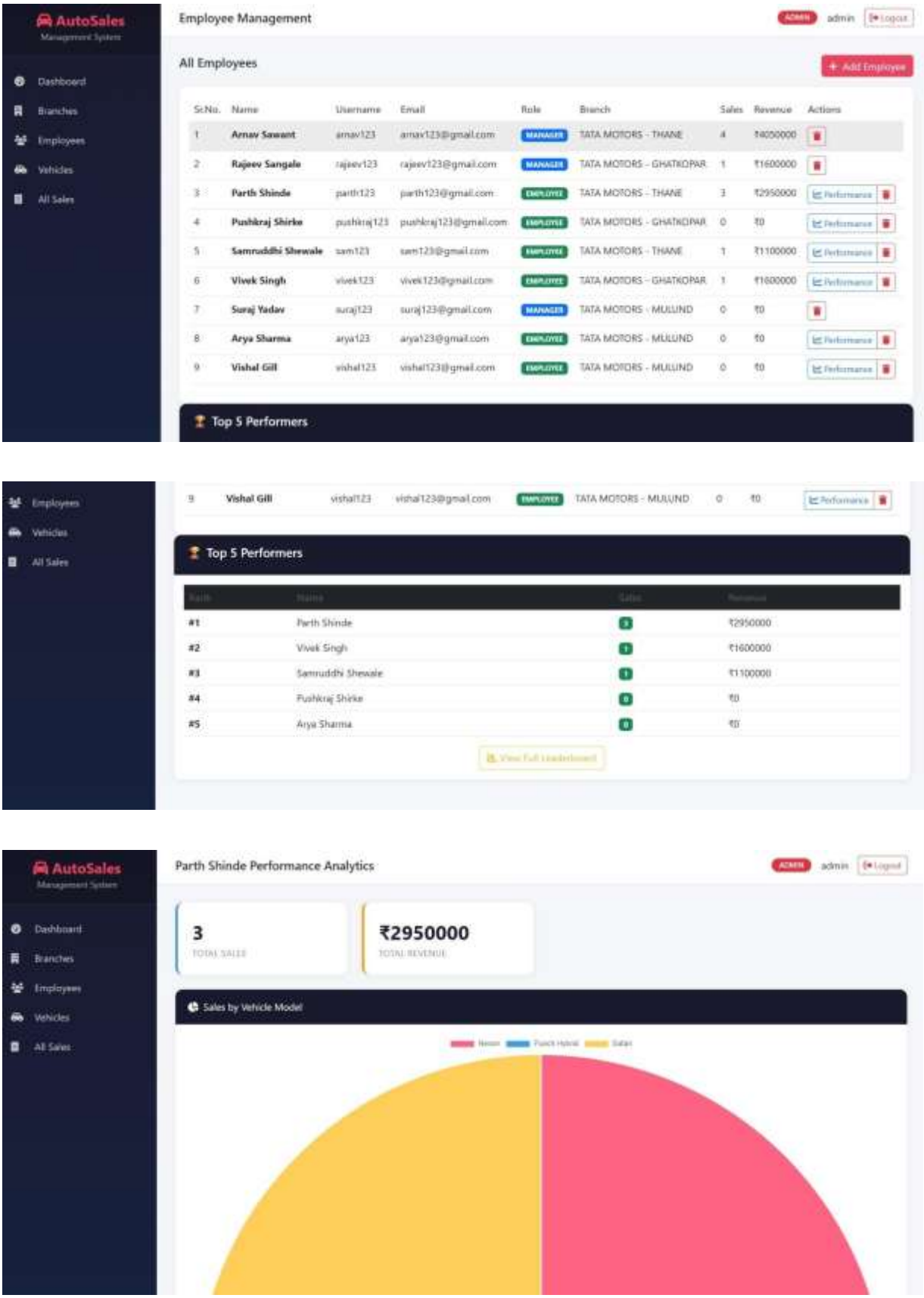

Figure 7.4 Employee Management and Performance Analysis

The Employee Management and Performance Analysis page allows the administrator to manage employee records and monitor their sales performance. It provides both detailed employee information and analytical insights, helping the admin evaluate employee contributions, identify top performers, and support better workforce management.

Key Features:

- **Employee Listing:** Displays all employees with details such as name, username, email, role, branch, sales count, and revenue generated.
- **Role Identification:** Clearly distinguishes managers and employees using role labels.
- **Performance Tracking:** Shows individual employee sales and revenue for performance evaluation.
- **Top Performers Section:** Highlights the highest-performing employees based on sales and revenue.
- **Performance Analytics:** Provides a dedicated performance page for each employee with total sales, total revenue, and vehicle-wise sales analysis.
- **Visual Reports:** Uses charts to present employee sales distribution in a clear and understandable format.
- **Employee Actions:** Includes options to add, edit, delete, and view detailed performance for employee records.
- **Administrative Control:** Helps the admin manage workforce data and make decisions based on employee productivity.

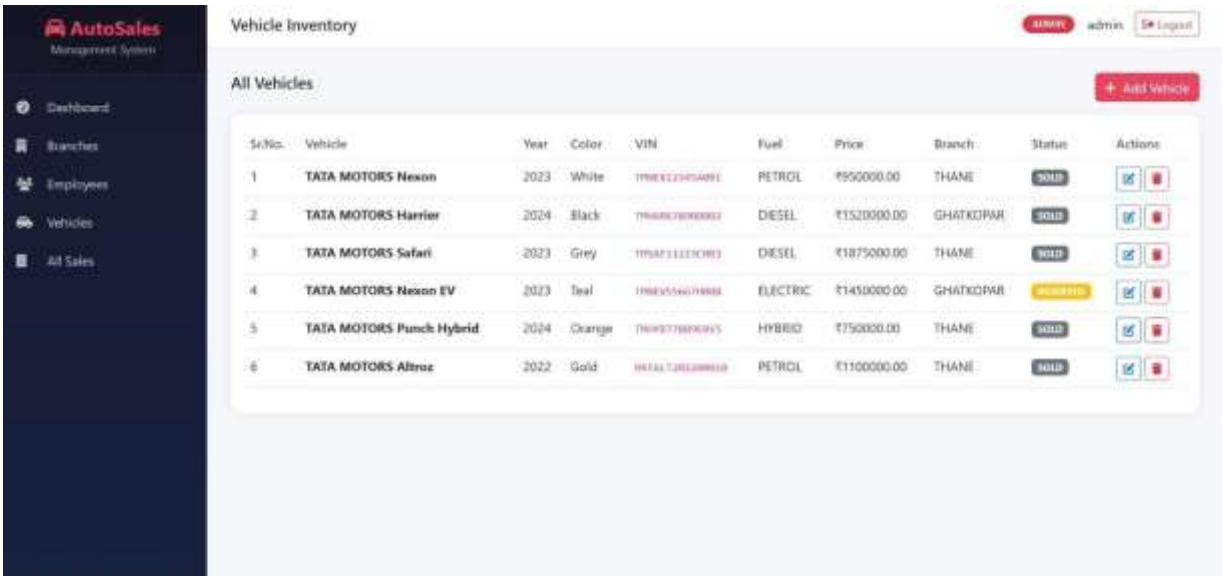

Figure 7.5 Vehicle Inventory

This system streamlines vehicle inventory operations, improves data organization, and provides administrators with efficient tools to manage their vehicle sales business across multiple locations.

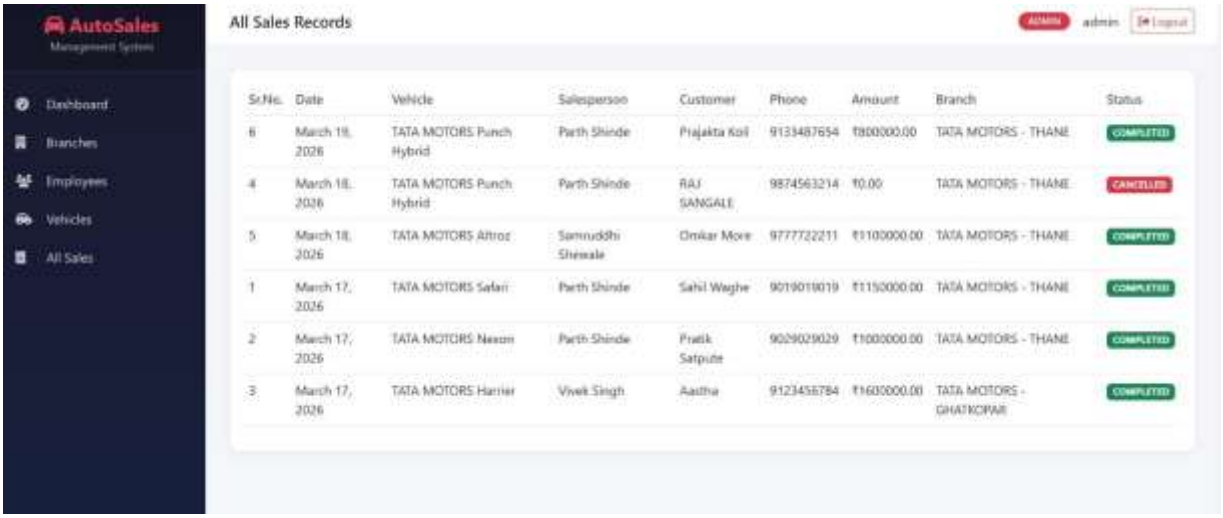

Figure 7.6 All Sales Record

This module streamlines sales operations, provides transparency in transaction history, and helps administrators monitor sales performance across multiple branches and salespeople.

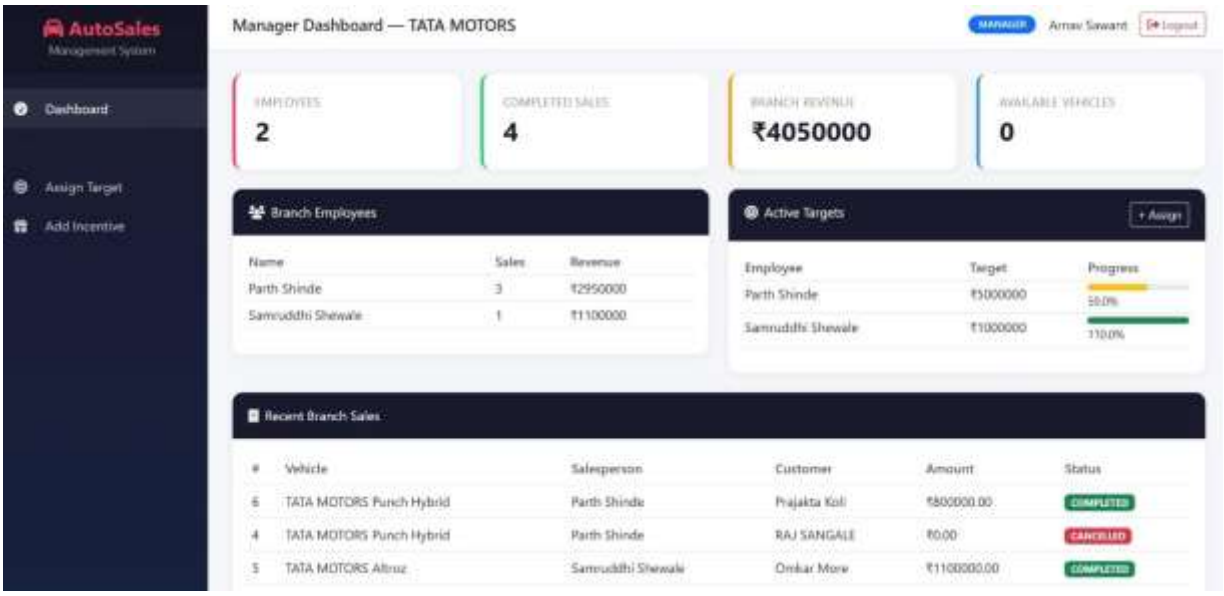

Figure 7.7 Managers Dashboard

The main dashboard provides key performance indicators (KPIs) and recent activity for the manager's branch:

- **Summary Cards:**
    - **Employees:** Total number of employees under this manager (e.g., 2).
    - **Completed Sales:** Total number of sales successfully completed (e.g., 4).
    - **Branch Revenue:** Total revenue generated by the branch (e.g., ₹4,050,000).
    - **Available Vehicles:** Current stock of vehicles ready for sale (e.g., 0, indicating low inventory or quick sales).
- **Branch Employees Section:** Lists individual employees with their respective sales count and total revenue generated.
- **Active Targets Section:** Displays assigned sales targets for each employee, showing their target amount and progress percentage towards achieving it.
- **Recent Branch Sales Section:** Shows a summary of the most recent sales transactions, including the vehicle, salesperson, customer, amount, and transaction status (e.g., COMPLETED, CANCELLED).

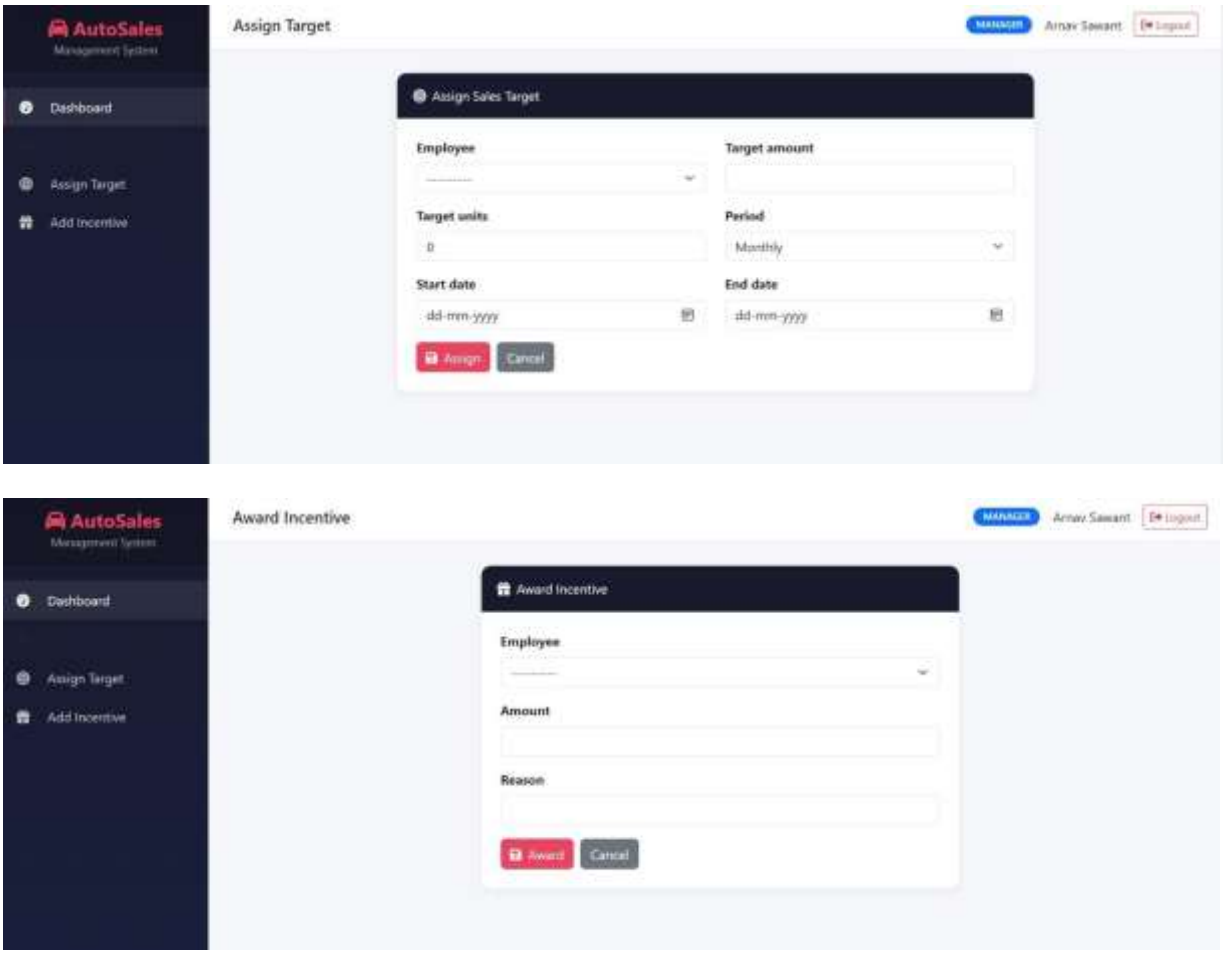

Figure 7.8 Assign Target and Award Incentive

Managers can set sales targets for their employees through a dedicated "Assign Target" interface. This functionality allows them to:

- Select an **Employee** from a dropdown list.
- Specify a **Target amount** (monetary value).
- Specify **Target units** (number of vehicles).
- Define the **Period** for the target (e.g., Monthly, Quarterly, Yearly).
- Set a **Start date** and **End date** for the target period.

Managers have option to award incentives to employees using the "Add Incentive" feature.

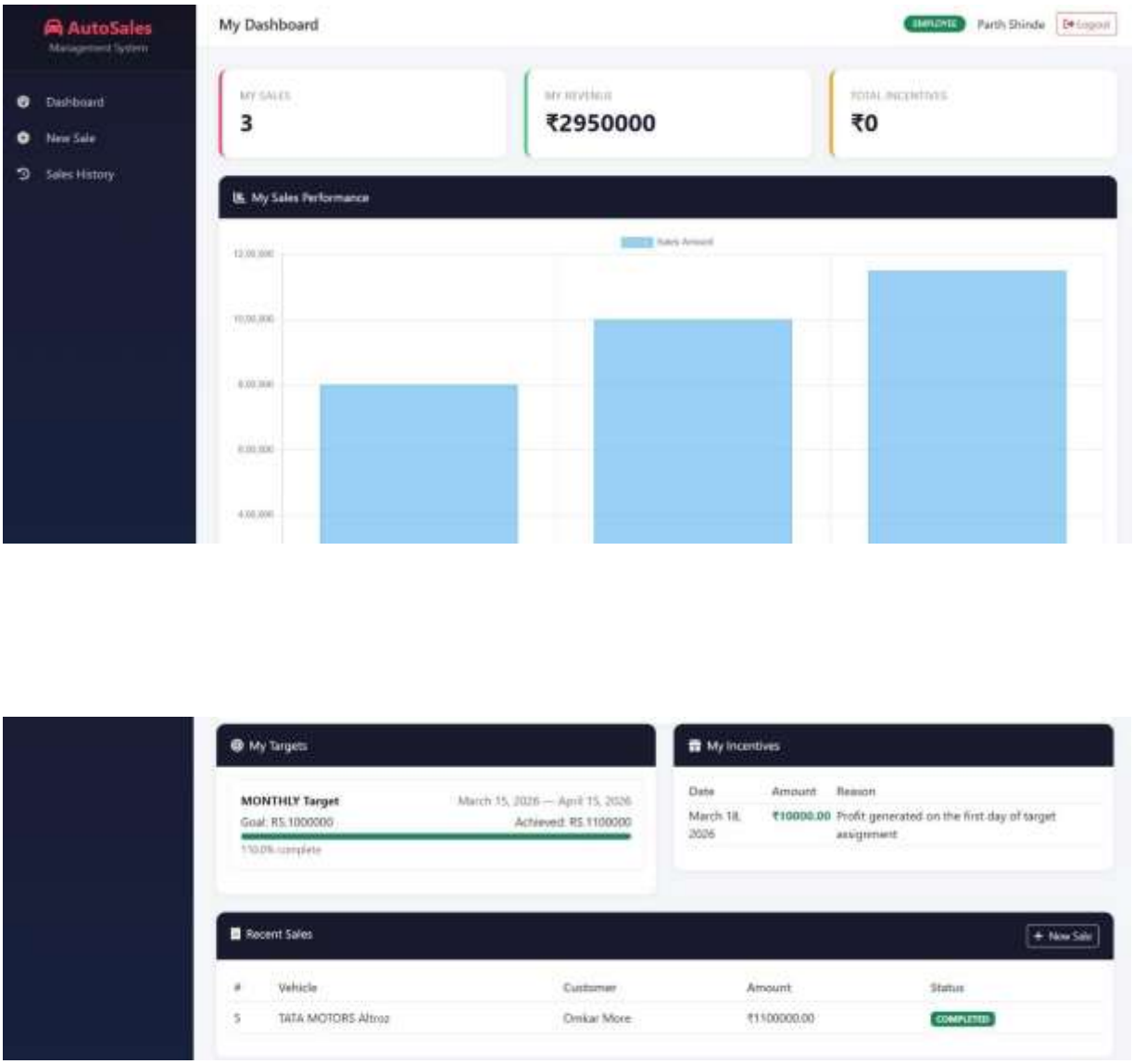

Figure 7.9 Employee Dashboard

The Employee Dashboard provides a personalized view of an employee's sales performance. It displays key metrics such as total sales, total revenue, and incentives earned. The dashboard includes a sales performance chart, assigned targets with progress tracking, incentive details, and recent sales records. It also allows employees to add new sales and review their sales history, helping them monitor performance and achieve targets efficiently.

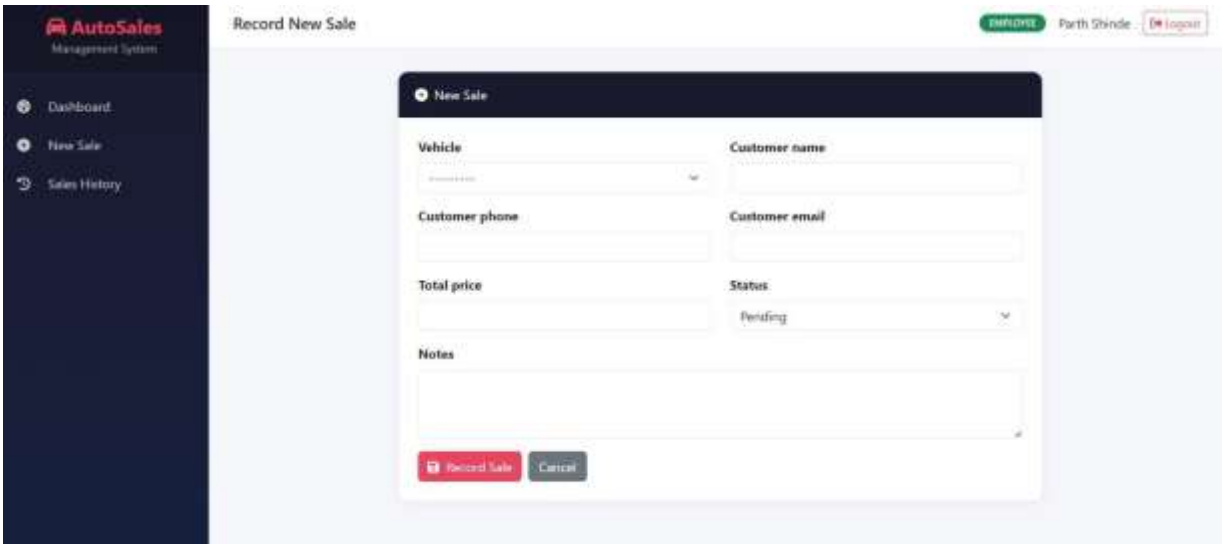

Figure 7.10 Record New Sale

The Record New Sale page allows employees to log a new vehicle sale by entering details such as vehicle name, customer name, phone, email, total price, sale status, and additional notes. The employee can submit the sale using the "Record Sale" button or cancel the entry.

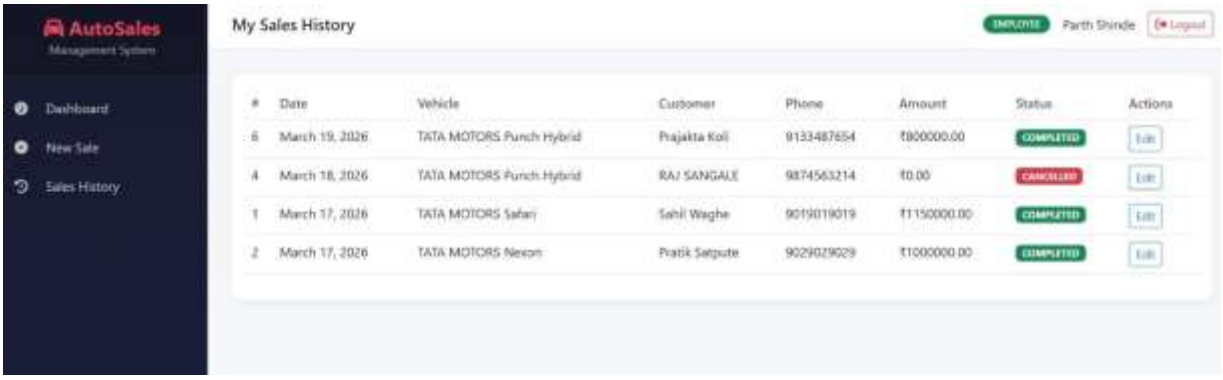

Figure 7.11 Sales History of Employee

The **Sales History** section displays a complete list of the employee's past sales, including sale date, vehicle details, customer information, amount, and transaction status. It also provides action options such as viewing or editing records, helping employees easily track and manage their sales activities

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The **Automobile Employee Management System** developed in this project successfully demonstrates an efficient and structured approach to managing employee-related activities within an automobile organization. The system integrates various modules such as employee management, branch management, vehicle tracking, sales monitoring, targets, and incentives into a unified platform.

The implementation of this system helps in reducing manual effort, minimizing errors, and improving overall operational efficiency. By digitizing records and automating processes, the system ensures better data accuracy, faster processing, and easy retrieval of information. It also provides analytical insights such as performance comparison, sales tracking, and employee productivity, which assist management in decision-making.

The use of modern technologies and frameworks ensures that the system is scalable, reliable, and user-friendly. The database design using ER modeling and the structured flow using DFD diagrams contribute to a well-organized architecture. Overall, the project fulfills its objective of creating a comprehensive solution for managing automobile employees and related operations effectively.

8.2 Future Scope

Although the system meets the current requirements, there are several enhancements that can be implemented in the future to further improve its functionality and usability:

- **Integration with AI/ML Models:**
  Advanced machine learning algorithms can be integrated to predict sales trends, employee performance, and customer behaviour for better strategic planning.
- **Mobile Application Development:**
  Developing an Android/iOS mobile application would allow employees and managers to access the system anytime and anywhere.
- **Real-Time Analytics Dashboard:**
  Implementing advanced dashboards with real-time data visualization can provide deeper insights into sales, targets, and performance metrics.
- **Cloud Deployment:**
  Hosting the system on cloud platforms like AWS or Azure would improve scalability, data security, and accessibility.
- **Online Payment Integration:**
  Implement secure payment gateways to allow customers to make vehicle or service payments directly through the system.
- **Smart Maintenance & Service Tracking:**
  Introduce modules to schedule, track, and notify customers about vehicle maintenance and service, including predictive maintenance alerts using AI.
- **Biometric or Attendance Integration:**
  Link with attendance systems for real-time employee tracking.
- **Automated Notification System:**
  Adding email/SMS notifications for targets, achievements, and important updates can improve communication within the organization.

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